

Parameters rule

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BZ Reaction & the Brusselator

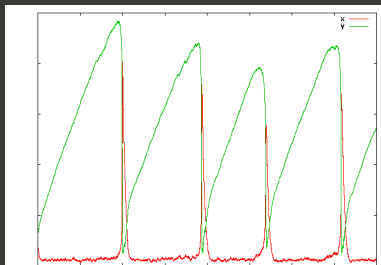


The **Brusselator**(JCP 46 1695) is a theoretical extremely simplified version (though physically unrealistic) of the **BZ** (Belousov-Zhabotinskii) Reaction now days recognized as being the **prototype for Chemical Oscillators**:



$$\begin{cases} \frac{\partial X}{\partial t} = k_1 A - k_2 B X + k_3 X^2 Y - k_4 X \\ \frac{\partial Y}{\partial t} = k_2 B X - k_3 X^2 Y \end{cases}$$

Parameters Influence

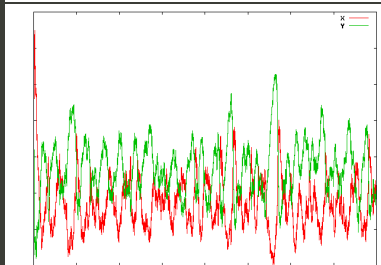


$$c_1 = 1$$

$$c_2 = 5 \cdot 10^{-3}$$

$$c_3 = 2.5 \cdot 10^{-5}$$

$$c_4 = 1.5$$



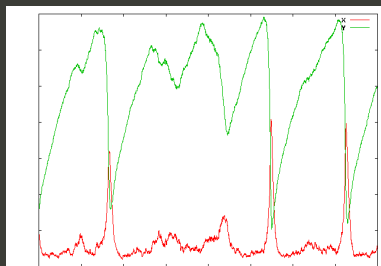
$$c_1 = 1$$

$$c_2 = 5 \cdot 10^{-3}$$

$$c_3 = 2.5 \cdot 10^{-4}$$

$$c_4 = 1.5$$

Parameters Influence

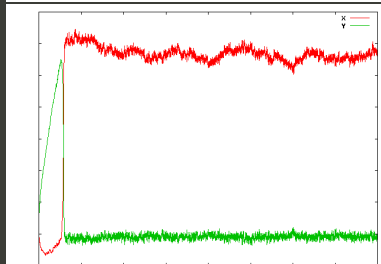


$$c_1 = 1$$

$$c_2 = 3.25 \cdot 10^{-3}$$

$$c_3 = 2.5 \cdot 10^{-5}$$

$$c_4 = 1.5$$



$$c_1 = 1$$

$$c_2 = 5 \cdot 10^{-3}$$

$$c_3 = 2.5 \cdot 10^{-5}$$

$$c_4 = 1.5 \cdot 10^{-1}$$

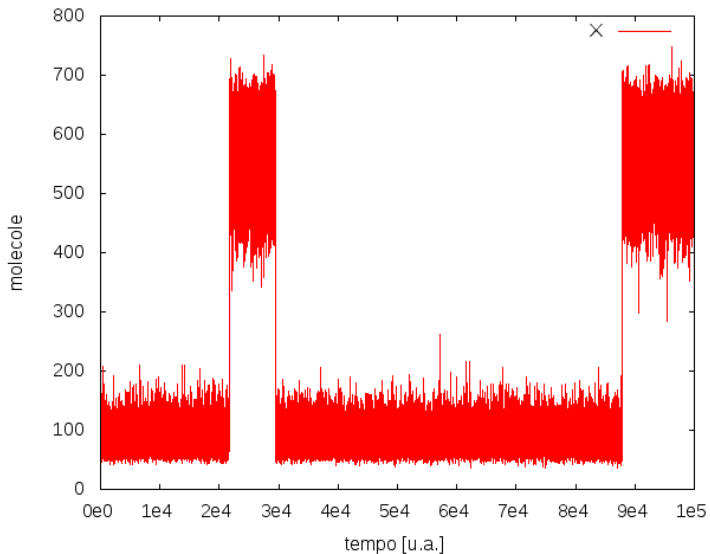
The Schlögl Model



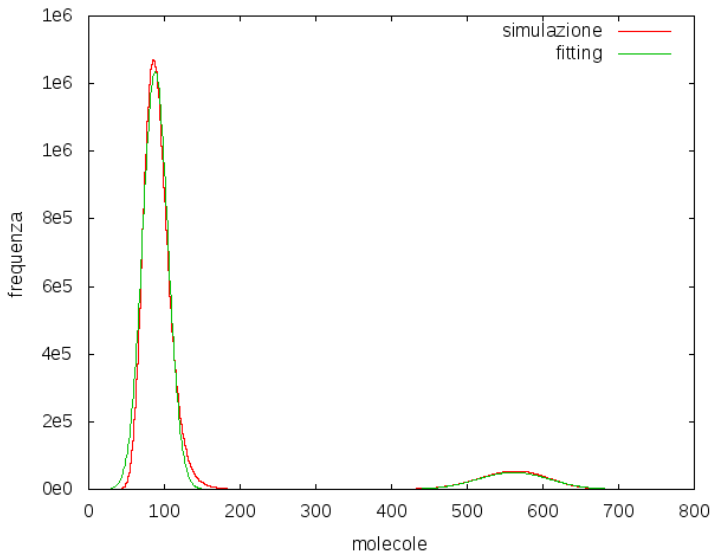
Kinetics:

$$\frac{\partial X}{\partial t} = k_1 A X^2 - k_2 X^3 + k_3 B - k_4 X$$

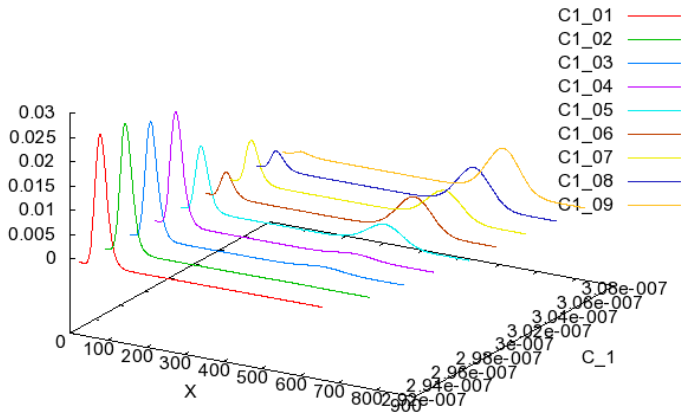
The Schlögl Model



The Schlögl Model



The Schlögl Model



The Schlögl Model

